Introduction to Economics

Class 8 — Surplus, efficiency, taxes and subsidies

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slide 1 of 28

Slide 1 Title

Block 3 begins. Blocks 1 and 2 asked how a market works: where it settles, which way it moves and how far. Block 3 asks whether the outcome is any good, and what a government does to it. The first half builds a way to measure what a market is worth to its buyers and its sellers, in euros, and shows why the market's own quantity makes that total as large as it can be. The second half puts a tax on the same market: who ends up paying it, where the money goes, and what a subsidy does; the thread through the class is a real VAT cut on bread, milk and eggs.

Quiz 5

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

quiz page, not numbered

Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet, write your name on the block, tear the strip off along the dashed line and hand in only the strip.

Plan for today

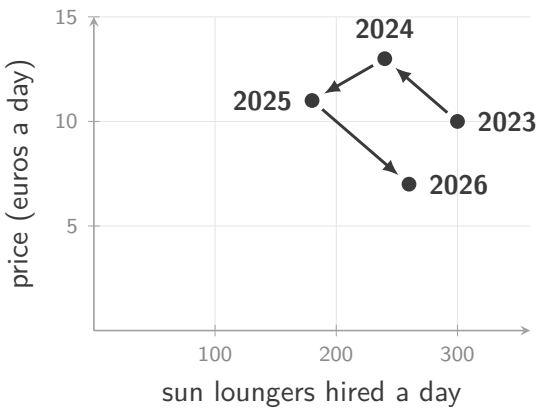
10:00	Quiz 5
10:10	Warm-up: which curve moved?
10:14	Bread, milk, eggs: VAT down to zero
10:20	What buyers gain: consumer surplus
10:26	What sellers gain: producer surplus
10:29	In pairs: menús del día
10:36	Total surplus: efficient, but fair?
10:44	A tax: a wedge between two prices
10:50	Break
11:00	Who pays the tax office; where the money goes
11:08	In pairs: a tax on menús
11:15	Who bears a tax: elasticity decides; in pairs
11:29	A subsidy; VAT; the bread answered
11:40	In one line; before Thursday 22 October

Slide 2 Plan for today

The class at a glance. The first half answers one question about Class 6's market: is its quantity any good? The tool is surplus, a way to put a buyer's gain and a seller's gain in euros; adding the two gives what the market is worth, and the market's own quantity turns out to make that as large as possible. The second half puts a tax on the same market: who bears it, where the money goes, and a subsidy as the same thing in reverse; a real VAT cut on basic food opens the class as a question and closes it as the answer.

Two halves of fifty minutes with a ten-minute break between them; two short warm-ups on earlier classes come first.

Warm-up: two dots — which curve moved?



Sun loungers on one beach, each August

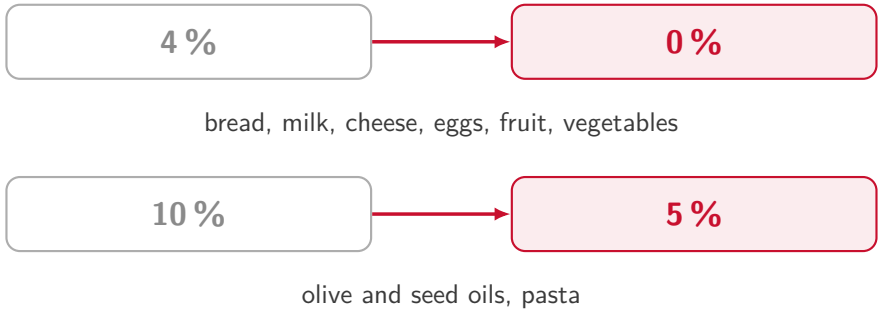
Each year only one curve moved. Which one, which way?

1. 2023 → 2024
   supply left: dearer, fewer
2. 2024 → 2025
   demand left: cheaper, fewer
3. 2025 → 2026
   supply right: cheaper, more

Warm-up 1 Two dots: which curve moved?

A warm-up on reading a market backwards: each dot is one August's equilibrium, and between two years only one curve moved, so the new dot lies on the curve that stayed put. From 2023 to 2024 loungers got dearer and fewer were hired, a move up along demand: supply shifted left. From 2024 to 2025 cheaper and fewer: demand shifted left; from 2025 to 2026 cheaper and more: supply shifted right. The rule: price and quantity moving the same way means demand moved, opposite ways means supply moved, so a smaller quantity on its own does not mean that demand fell.

1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %



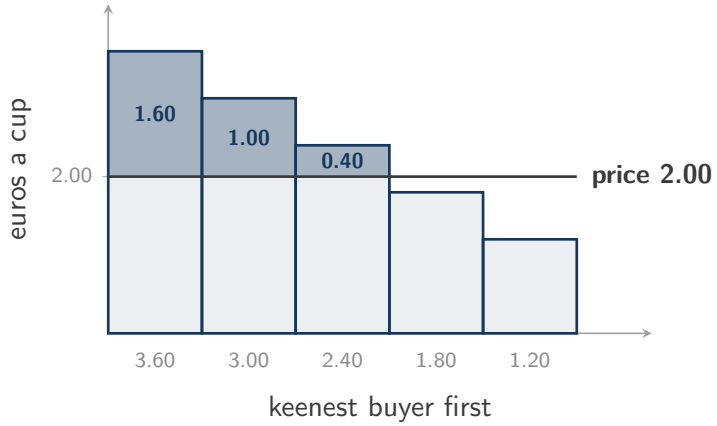
A 1.04 € loaf: 4 cents of VAT.
How many reached the shopper?

Real Decreto-ley 20/2022 (BOE, 28 December 2022), in force 1 January 2023.

Slide 4 1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %

The hook, from the official gazette. From the start of 2023 the government took the VAT off the most basic foods and halved it on oils and pasta, to fight food inflation. On a loaf that cost 1.04, four cents were VAT; after the cut the shop handed nothing to the tax office for it. Whether the loaf then cost 1.00, or the shop kept some of the four cents, is the question the second half answers; consumer groups at the time said many shops had not passed it on, and the papers ran the story. Keep the question in mind while the tools are built: it needs a way to measure who gains from a market.

Willingness to pay: each step of demand is one buyer



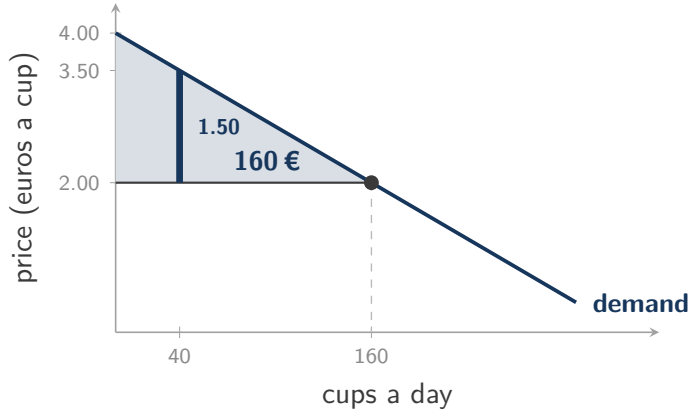
Consumer surplus = willingness to pay − price: 1.60 + 1.00 + 0.40 = 3.00 €.

Slide 5 Willingness to pay: each step of demand is one buyer

Five buyers at one counter, each with the most they would pay this morning; that maximum is the willingness to pay, how much a buyer values the good, in euros. Lined up keenest first, the five steps are a demand curve: at any price, the buyers whose step rises above it buy, the others do not. What a buyer would have paid minus what she actually pays is her gain from the trade, her consumer surplus; the two whose steps lie below the price do not buy and gain nothing.

Try it yourself: drop the price to 1.50 and recount. The surplus grows in two ways, because the old buyers pay less and a new buyer joins.

Consumer surplus: under demand, above the price — 160 € a day



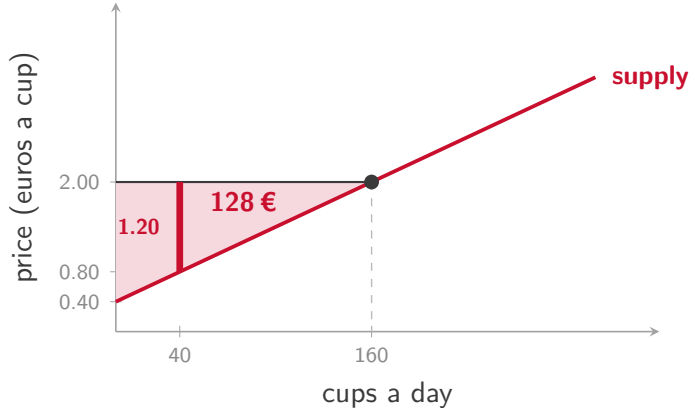
½ × 160 cups × 2.00 € = 160 euros a day: every buyer's bargain, added up.

Slide 6 Consumer surplus: under demand, above the price, 160 € a day

The staircase with 320 steps, each one cup wide, is Class 5's straight line, and its height at any cup is the willingness to pay of whoever buys that cup. The bar at the 40th cup is one buyer's gain, 3.50 minus the price; the buyer of the 160th cup pays exactly what it is worth to him and gains nothing: the marginal buyer, the first to leave if the price rose. Adding every bar gives the triangle, and a triangle's area is half the base times the height; forgetting the half is the usual slip.

The other slip is to take the box below the price, 2.00 times 160 cups, for the surplus. That box is what buyers spend, Class 7's revenue; the surplus is what they would have paid on top of it, above the price.

Producer surplus: above supply, below the price — 128 € a day



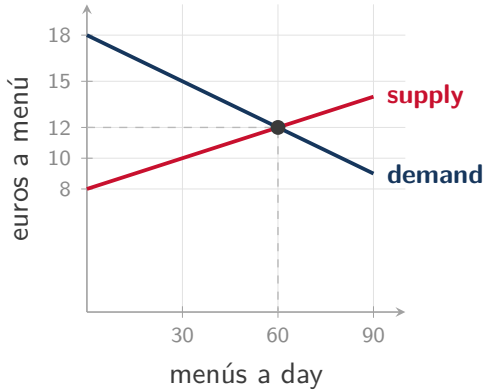
½ × 160 cups × 1.60 € = 128 euros a day: what the bars get above their costs.

Slide 7 Producer surplus: above supply, below the price, 128 € a day

The mirror image for sellers. A bar makes one more cup only if the price covers that cup's cost, everything the bar gives up to make it, the owner's own time included; so the height of supply at any cup is the cost of the bar that makes it, and the bar making the 160th cup is the marginal seller, the first to stop if the price fell. The price received minus the cost, added over all the cups, is the triangle above supply and below the price; its height is the price minus 0.40, the lowest cost on the line.

Producer surplus is close to profit but not the same: the bar's rent is paid whether or not this cup is made. Costs are treated properly in Class 11.

In pairs, four minutes: read the surplus off the graph



Menús del día on one street

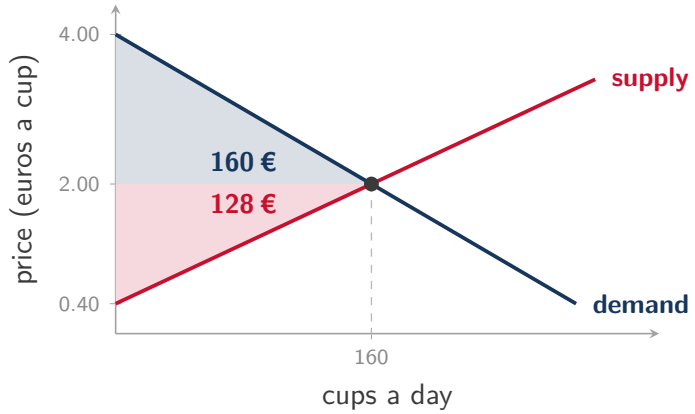
1. Consumer surplus at 12 €?
   buyers get 180 € a day
2. Producer surplus at 12 €?
   the bars get 120 € a day
3. The 30th menú: worth? costs?
   worth 15, costs 10: 5 € of surplus

Slide 8 In pairs, four minutes: read the surplus off the graph

Exercise 1, on a different market, so that the method rather than the coffee numbers is what you practise. The consumer surplus triangle runs from the price up to where demand meets the price axis, 18, and the producer surplus triangle from the price down to where supply starts, 8; both have the 60 menús as their base, and both need the half. Item 3 reads one unit instead of an area: go up from the 30th menú to demand for what its buyer would pay and to supply for what the bar that makes it needs; the gap between the two is that menú's own surplus, and the price splits it between the two sides.

Three traps: 720 is 60 times 12, what buyers spend, not what they gain; 360 or 240 means the half was forgotten; and 12 is the price, not what the 30th menú is worth to its buyer.

Total surplus: buyers' 160 plus the bars' 128 — 288 euros a day

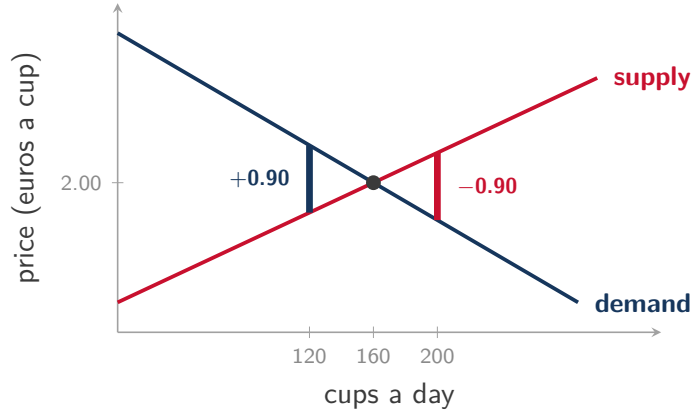


Total surplus = value to buyers − cost to sellers = 480 − 192 = 288 euros a day.

Slide 9 Total surplus: buyers' 160 plus the bars' 128, 288 euros a day

The two triangles together fill the area between the curves up to the market quantity, and that area is what the market is worth: the value of the cups to those who drink them, the whole area under demand, minus what they cost the bars that make them, the area under supply. The price does not appear in the total, because what buyers pay is exactly what the bars receive; a higher price only moves surplus from the blue triangle to the red one. That is why the next question is about the quantity, not the price.

Stop at 160: before it, cups worth more than they cost; after it, less



Below 160: surplus left on the table. Above: cups that cost more than they are worth.

Slide 10 Stop at 160: before it, cups worth more than they cost; after it, less

Why the market quantity is the right one, read one cup at a time. For any cup to the left of the crossing the demand line sits above the supply line: someone values that cup at more than it costs a bar to make, so not making it throws surplus away, 0.90 of it at the 120th cup. For any cup to the right, supply sits above demand: making it uses up more than it is worth. Only at the crossing is the next cup worth exactly its cost, so no other quantity gives a larger total. The 0.90 gap at 120 cups comes back three slides on, as a tax.

Just right: the market's 160 cups give the largest total surplus

- The cups go to those who value them most
- They are made by the bars that make them cheapest
- Every cup worth more than its cost, and no other

That is what economists call **efficient**, as long as no trade affects others (Class 10).

Slide 11 Just right: the market's 160 cups give the largest total surplus

Slide 3's answer. The three lines are the reading's three reasons, and each one is a sorting: at 2.00 the cups go exactly to the people who value one at 2.00 or more, so nobody who values a cup less takes it from someone who values it more; the bars that make them are exactly those that can do so for 2.00 or less; and slide 10 settled the quantity. An outcome that makes total surplus as large as it can be is called efficient, and the market reaches it with nobody planning it.

Two conditions sit behind this: everyone is a price taker, and the trade affects nobody outside it. When a bar's terrace keeps the street awake, that cost is on no curve; that is Class 10.

Efficient is not the same as fair

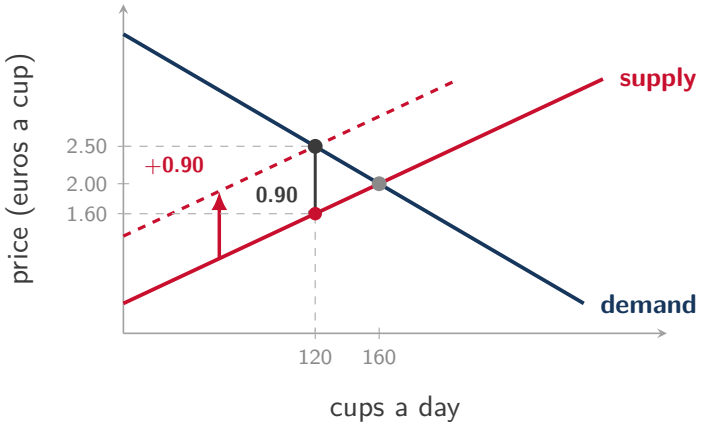
- Efficiency: is the cake as big as it can be?
- Equity: is it shared fairly?
- Willingness to pay depends on income

Efficiency is a positive question; fairness a normative one (Class 4).

Slide 12 Efficient is not the same as fair

Two different questions about the same cake. Whether the cake is as big as it can be is measurable, so efficiency is a positive claim; whether it is cut fairly is a judgement, so equity is normative, Class 4's distinction. The third line says why the two can come apart: "values most" means "would pay most", and what someone can pay depends on their income. A tourist outbidding a young local for the last flat in Gràcia is efficient by this measure and may still strike you as unfair; economics measures the cake, you judge the slices.

A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



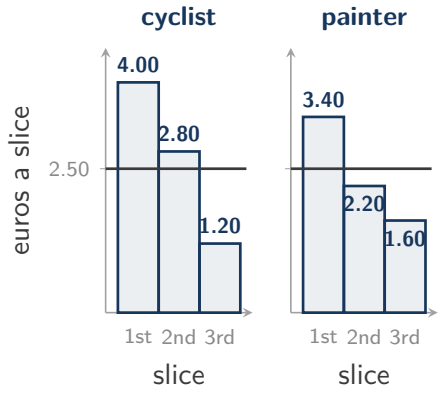
Buyers pay 0.50 more, the bars keep 0.40 less: a tax is a wedge between two prices.

Slide 13 A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60

A tax on every cup sold, handed in by the bars, is an indirect tax: a tax on spending rather than on income. Supply moves up by exactly the tax, because a bar now needs 0.90 more from the customer before any given cup is worth making; it is Class 6's supply shift, drawn again. At the new crossing the buyers' price is read on the shifted line and the bars' price, what they keep after handing over the tax, on the old line directly below it: the tax opens a wedge between two prices, and the market shrinks to the quantity where that wedge just fits between the curves, slide 10's gap.

Neither side pays the whole 0.90: the price rises by less than the tax, and the bars bear part of it although they are the ones who hand it in. Why buyers carry a little more than half is Class 7's elasticities, made into a rule after the break.

Market demand from two staircases (optional)



Pizza slices: two buyers

1. At 2.50: slices bought in all?
   3: cyclist 2 + painter 1
2. Their consumer surplus at 2.50?
   1.50 + 0.30 + 0.90 = 2.70 €
3. Slices bought in all at 2.00?
   4 slices: 2 + 2

Slide 14 Market demand from two staircases (optional)

An optional exercise on how a market demand curve is built. Each staircase is one buyer's willingness to pay for a first, second and third slice, as on slide 5; at a given price each buyer takes every slice worth at least that price, and the market buys the sum of those quantities at that same price: at 2.50 the cyclist takes two slices and the painter one, three in all, and at 2.00 four. The consumer surplus adds, slice by slice, what each slice bought is worth minus the price: $1.50 + 0.30 + 0.90 = 2.70$ euros at 2.50. The market staircase is the six steps sorted from the highest, the two staircases pushed together sideways.

The trap: adding the prices two buyers would pay for their first slices stacks the curves upward and gives no point on the market curve.

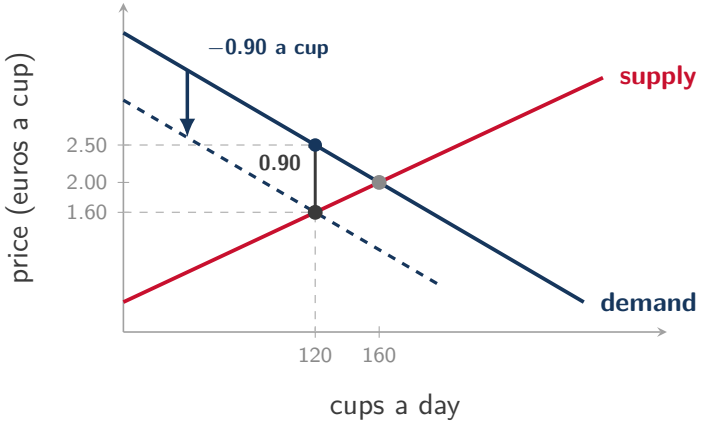
Break

Back at 11:00

Slide 15 Break

Ten minutes. After it: who hands the tax to the tax office, and where the money goes.

Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

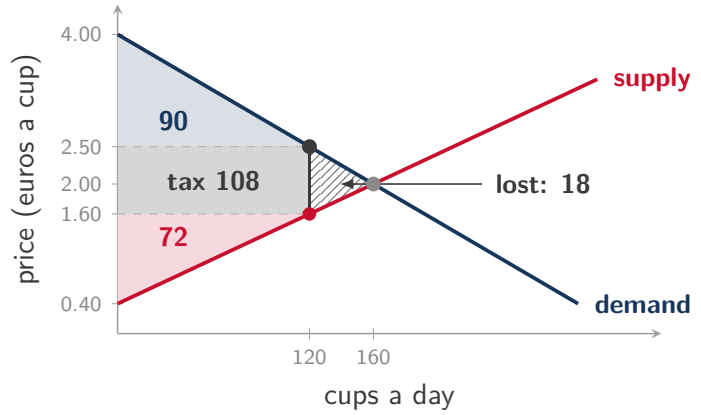


Who hands the 0.90 to the tax office does not matter: the wedge does.

Slide 16 Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

The same tax, collected from the other side. If customers pay the 0.90 themselves, a cup with a counter price P costs them P plus 0.90, so what each buyer is willing to pay at the counter falls by exactly the tax: demand moves down by 0.90 instead of supply moving up. The new crossing gives the counter price, which is what the bars keep; adding the tax back gives what buyers pay in all, and the three numbers are slide 13's. The law chooses who hands the money over; it cannot choose who bears it, because the wedge between the two prices is the same whichever curve moves. Most governments collect from sellers simply because there are fewer of them to check.

With the tax: buyers 90, bars 72, the state 108 — and 18 lost



Buyers lose 70, the bars 56; the state gets 108. The lost 18: the **deadweight loss**, Thursday.

Slide 17 With the tax: buyers 90, bars 72, the state 108, and 18 lost

Slide 9's 288, after the tax. The buyers' triangle now starts at the higher price and the bars' triangle ends at the lower one, so both shrink; the rectangle between the two prices, as tall as the tax and as wide as the cups still sold, is what the state collects. The three areas add up to 270, not 288: the missing 18 went to nobody, because the forty cups between 120 and 160 are no longer made, and slide 10 showed that each of them was worth more than it cost. That triangle is the deadweight loss, the fall in total surplus that a tax causes; today it gets its name and its picture, and Thursday's reading is about it.

The trap is to count the state's 108 as a loss: that money pays for something, and the loss to everyone together is only the 18.

In pairs, four minutes: a tax of 2.50 on every menú

Same street, same menús

menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

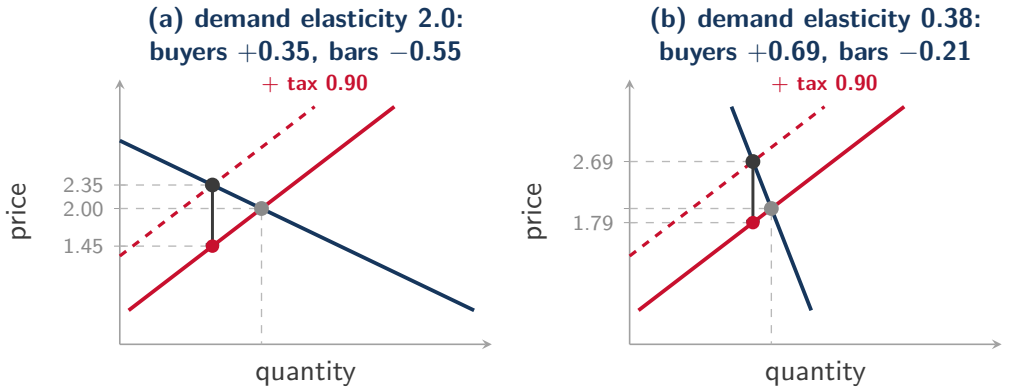
1. No tax: price and menús?
   12.00 and 60 menús
2. Tax 2.50: menús, buyers pay, bars keep?
   45 menús: buyers 13.50, bars 11.00
3. Tax collected a day?
   2.50 × 45 = 112.50 €
4. Who bears more of the 2.50?
   buyers: 1.50 of the 2.50

Slide 18 In pairs, four minutes: a tax of 2.50 on every menú

Exercise 2 is slide 8's street as a table: "worth" is the height of demand at that quantity and "costs" the height of supply, so the table is the two curves without a drawing. Without a tax the market stops where the two columns are equal. With a tax it stops where the gap between worth and cost is exactly the tax, because that gap is the wedge: the worth column is then what buyers pay and the cost column what the bars keep.

The tax is collected on the menús still sold, not on the old quantity; that is the trap in item 3. Item 4 compares each side's new price with the no-tax price: the side that moved more bears more, and the next slide says why.

Class 7's markets, taxed 0.90: who cannot walk away pays more



Buyers who can switch leave most of the tax to the bars; buyers who cannot pay most of it.

Slide 19 Class 7's markets, taxed 0.90: who cannot walk away pays more

Class 7's last figure, read again. The dashed line is the same supply shift as then, and today it has a cause: a tax of 0.90 a cup, with the supply curve identical in both panels. In (a) buyers can switch, so when the tax arrives they leave: the price they pay rises only a little and the bars absorb most of the 0.90. In (b) buyers have few substitutes, so most stay and pay: the price rises by most of the tax and the bars keep almost what they kept before. Nothing on the sellers' side differs between the panels, so the whole difference in who bears the tax comes from Class 7's elasticity of demand.

Tax incidence: the less elastic side bears more of the tax

	most of the tax falls on
demand less elastic	buyers
supply less elastic	sellers
equally elastic	half each

The law says who pays the tax office; elasticities say who bears the tax.

Slide 20 Tax incidence: the less elastic side bears more of the tax

The word for how the burden of a tax is shared is tax incidence, and the table is the rule the last two slides illustrated. An elasticity measures how easily a side can leave the market when the deal gets worse; the side with fewer alternatives stays, and whoever stays bears the tax. The coffee tax, the menú tax and the two panels all fit the rule once the two elasticities are known. Who is named in the law is a separate question altogether, as slide 16 showed.

In pairs, three minutes: who bears most of the tax?

Buyers or sellers — and why?

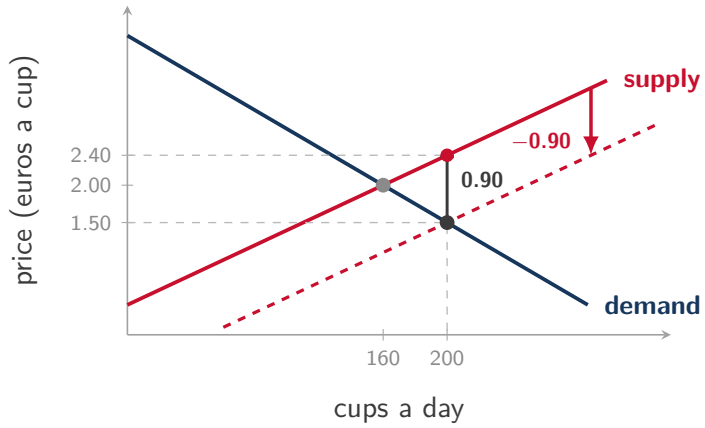
1. Cigarettes: demand elasticity 0.4, supply 2.0
   buyers: smokers stay
2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
   sellers: the umbrellas are there anyway
3. Tax 1.00; buyers' price 5.00 → 5.80. Sellers keep? Less elastic side?
   sellers keep 4.80; buyers bear 0.80: demand less elastic
4. The same tax, paid by the buyers: the seller's price, tax not included?
   4.80, tax not included; buyers pay 5.80 in all

Slide 21 In pairs, three minutes: who bears most of the tax?

Exercise 3 applies the rule, twice from elasticities and twice from prices. In items 1 and 2 compare the two numbers: the smaller elasticity marks the side that cannot leave, and that side bears most; smokers do not quit when cigarettes get dearer, while this Sunday's umbrellas are on the beach whatever the price and a bather can sit two metres away. In item 3 the elasticities are not given but the prices are: subtract the tax from what buyers pay to get what sellers keep, and compare each side's move with the old price; the side that moved more is the less elastic one. Item 4 is slide 16 again: when buyers hand over the tax, the seller's price, tax not included, is what the seller keeps, and the buyer adds the tax on top, with the same split.

Two traps: "the side the law names pays", and "the seller's price stays the same and buyers pay the full tax on top", which forgets that the market resettles.

A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



The tax reversed: buyers save 0.50, the bars gain 0.40.
The state pays $0.90 \times 200 = 180\,€$ a day.

Slide 22 A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40

A subsidy is a tax with the sign changed: the state pays the bars a fixed sum for every cup, so a bar needs that much less from the customer and supply moves down by the subsidy. The market grows past 160, buyers pay less and the bars receive more, and the gap between the two prices is again 0.90, now paid by the state rather than collected by it. The split is the same as under the tax, because the same two elasticities decide it: a subsidy paid to sellers does not stay with the sellers.

The bill is the subsidy times every cup sold, including the cups that were sold anyway; and the extra cups past 160 cost more than they are worth, slide 10's 200th cup, so a subsidy has a lost triangle of its own. Class 5's T-usual fare is a subsidy of this kind.

Per unit or per cent: VAT is a percentage of the price

- A fixed sum per unit: a **specific tax** (0.90 a cup)
- A share of the price: **ad valorem** (VAT)
- Spain's VAT: 21 %, 10 %, 4 %
- A loaf: 1.00 + 4 % VAT = 1.04 €

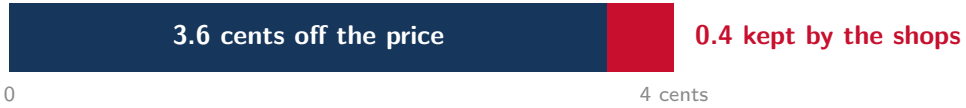
Either way a wedge; the elasticities decide who bears it.

Slide 23 Per unit or per cent: VAT is a percentage of the price

Two ways to write a tax. A specific tax is a fixed sum per unit, like today's 0.90 a cup or the tax on a litre of fuel; an ad valorem tax is a share of the price, so its wedge is wider where the price is higher, which is why the reading draws supply turning rather than shifting. Spain's three rates put most goods at the top one, restaurants and some foods in the middle, and the basics at the bottom, where slide 4's loaf sat. The rule of slide 20 does not care which kind of tax it is: either way there is a wedge, and the elasticities decide who bears it. The next slide takes that rule back to the bread.

The VAT cut on basic foods: about 90 % reached shop prices

The 4 cents of VAT on a 1.04 € loaf, split at 90 % (the figure for all the cut foods)



Inelastic buyers bore most of the VAT, so they got most of the cut.

January–February 2023, large supermarkets: over 90 % (EsadeEcPol, Brief 36, 30 March 2023).

One large chain, all foods with the cut: about 90 % by week 2 (Banco de España, Working Paper 2417, 2024).

Slide 24 The VAT cut on basic foods: about 90 % reached shop prices

Slide 4's answer, and the class's rule confirmed on real prices. Two independent studies compared the foods that lost their VAT with similar foods that kept it, in supermarket prices, for all the foods with the cut taken together: EsadeEcPol, Brief 36, found over 90 per cent of the cut in shop prices in January and February 2023, and the Banco de España, Working Paper 2417, found about 90 per cent by the second week in one large chain's prices. Slide 20 predicted it: demand for basic foods is inelastic, Class 7's necessities, while a supermarket can stock more at about the same cost, so supply is elastic; the less elastic side bore the VAT and therefore got the cut.

Many shoppers felt nothing because food prices were still rising for other reasons in 2023; the honest comparison is with foods that kept their VAT, not with last year's price. Slide 12 again: half of the money the state gave up went to the best-off 40 per cent of households, who spend more euros on food; whether that is fair is a normative question.

Which curve moves? (optional)

Cinema tickets in one city

1. A tax of 1.00 a ticket, paid by the cinemas: which curve shifts?
   supply, up by 1.00
2. Tickets get dearer: does demand shift too?
   no: a move along demand
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
   shifts left: a cheaper substitute
4. And the ticket price, and the tickets sold?
   both fall: a move along supply

Slide 25 Which curve moves? (optional)

An optional exercise on the difference between a shift of a curve and a move along it. A tax collected from the cinemas raises what each cinema needs per ticket, so supply moves up by the tax; the dearer ticket then moves buyers along their unchanged demand curve, because a good's own price never shifts its own curve. A subsidy that makes streaming cheaper changes the price of a substitute, so fewer tickets are wanted at every price and ticket demand shifts left; the cinemas' supply does not move, so the price and the number of tickets both fall along it. One event, one shift; the price change is always a move along the other curve.

Who gains from trade? (optional)

Hours for one unit

	1 kg of cheese	1 bottle of wine
valley farm	2	1
hill farm	3	6

1. Who should make the cheese, and why?
   hill: a kg costs it 0.5 bottle, the valley 2
2. At 1 bottle a kg: each farm's gain?
   hill +0.5 bottle, valley +1: 1.5 shared
3. Hill farm at 4 h a kg, 2 h a bottle: any gain?
   no gain: both give up 2 bottles a kg

Slide 26 Who gains from trade? (optional)

An optional exercise that joins Block 1's comparative advantage to today's surplus. What a kilo of cheese costs is what else the same hours could make: 3 hours on the hill farm would make half a bottle of wine, 2 hours in the valley two bottles, so the hill farm should make the cheese although the valley is faster at both. At 1 bottle a kilo the hill farm gains half a bottle on each kilo and the valley one bottle: the gain, $2 - 0.5 = 1.5$ bottles a kilo, is shared like a surplus, and the price decides the split, as it did for slide 8's 30th menú; any price between 0.5 and 2 leaves both better off. If both farms gave up the same 2 bottles for a kilo there would be nothing to gain: trade pays only when opportunity costs differ.

Surplus: value minus price for buyers, price minus cost for sellers.

The market's quantity makes total surplus as large as it can be.

A tax or a subsidy is a wedge; the less elastic side feels most of it.
Deadweight loss and price controls: Thursday.

Slide 27 In one line

The class in four lines, to read slowly: line two is slide 11's answer (just right), line three is the rule of slide 20, and line four names Thursday.

Before Thursday 22 October

- Thursday: price controls; international trade and tariffs
- Read Mankiw & Taylor: Ch. 7, pp. 159–163 and 172–178; Ch. 17, pp. 380–390
- Problem set 1: due Thursday 22 October, on paper, in class
- Practise in Sextant: the units the course page lists

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Slide 28 Before Thursday 22 October

Thursday 22 October: what happens when the government sets the price itself, rent caps and price floors, then the deadweight loss properly, and international trade and tariffs. Reading: Mankiw & Taylor chapter 7, pages 159 to 163 and 172 to 178, and chapter 17, pages 380 to 390. Problem set 1, handed out on 15 October, is due on Thursday 22 October, on paper, in class, one per person. Practise in Sextant with the units the course page lists.

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